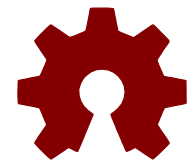
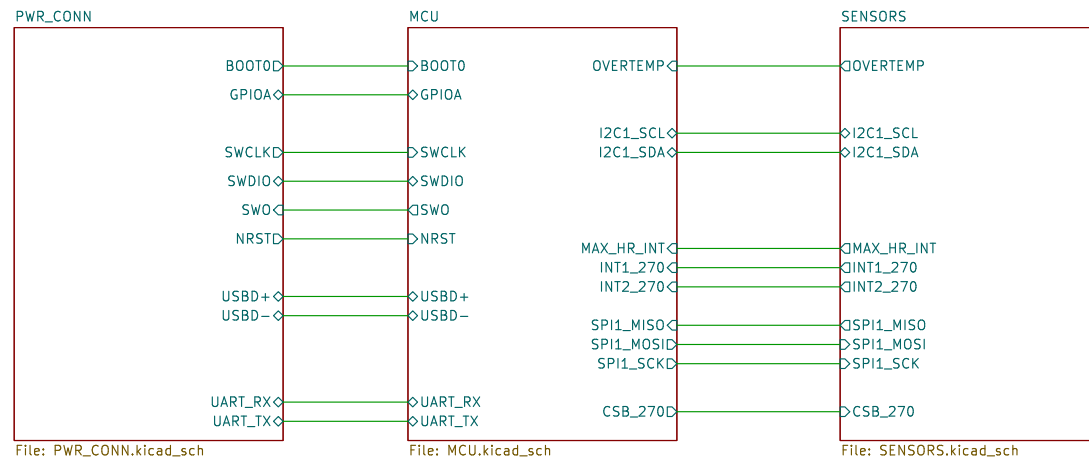


ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC



STM32 SENSOR EVALUATION BOARD



OPEN TO PUBLIC

UNDER REVISION
DO NOT MANUFACTURE
NOT FOR MEDICAL OR FOR USE IN
CRITICAL APPLICATIONS
OPEN TO PUBLIC
NOT RELATED TO ANY
ORGANIZATION
USE AT YOUR OWN RISK

Future Revisions:
-BLE/WiFi,4G LTE
-RTC, LCD, Crypto Module
-QSPI Mass storage
-BMS pseudo-MPPT
-Safety features, PMOS RPP,
PTC fusing, PMIC, Ideal diodes
-STM32u585 -> Low power ics

ENGINEERING PROTOTYPE V1.0.0
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Sheet: / File: STM32_SNSR_TESTS.kicad_sch		
Title: Root Sheet / STM32_SNSR_TESTS		
Size: A4	Date: 2026-01-12	Rev: 1.0.0
KiCad E.D.A. 9.0.7	Id: 1/4	

The diagram shows the SWD interface connection. The J2 header (SWD_02x05) is connected to the ESDA6V1-55C6 chip (D1) as follows:

- Pin 1 of J2 is connected to +3V3.
- Pin 2 of J2 is connected to R2 (0R) → ◇SWDIO.
- Pin 3 of J2 is connected to R3 (0R) → ◇SWCLK.
- Pin 4 of J2 is connected to R4 (0R) → ◇SWD.
- Pin 5 of J2 is connected to R5 (0R) → ◇NRST.
- Pin 6 of J2 is connected to Pin 1 of D1.
- Pin 7 of J2 is connected to Pin 2 of D1.
- Pin 8 of J2 is connected to Pin 3 of D1.
- Pin 9 of J2 is connected to Pin 4 of D1.
- Pin 10 of J2 is connected to Pin 5 of D1.

The ESDA6V1-55C6 chip has pins 1 through 5 labeled on its top edge and pin 2 labeled on its bottom edge. The chip is labeled D1 and ESDA6V1-55C6.

ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC

Title: PWR_CONN

Size: A4	Date:
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KiCad E.D.A. 9.0.7

Date: 2026-01-12

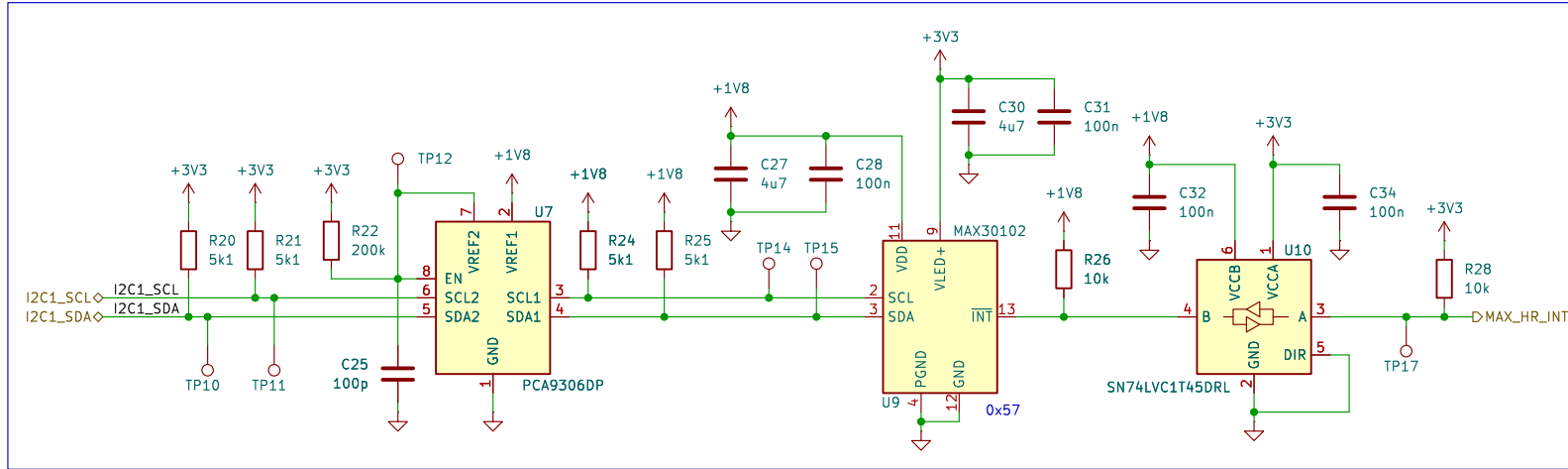
Rev: 1.0.0

Id: 2/4

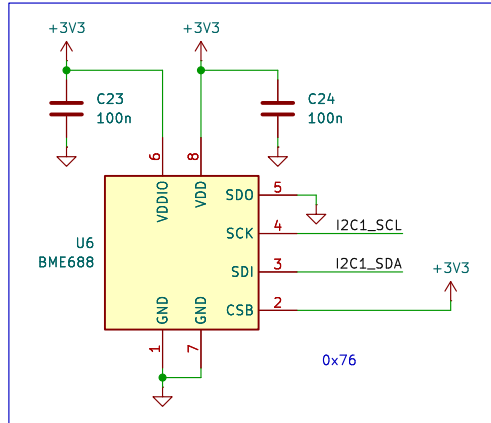
Schematic diagram of the ATECC608B-SSHDA module. The module is represented by a yellow box with pins 1-6. Pin 1 is GND, connected to a 0x60 label. Pin 2 is VCC, connected to a +3V3 supply. Pin 3 is SDA, connected to I2C1_SDA (pin 5). Pin 4 is SCL, connected to I2C1_SCL (pin 6). A capacitor C35 (100nF) is connected between VCC and GND.

Id: 3/4

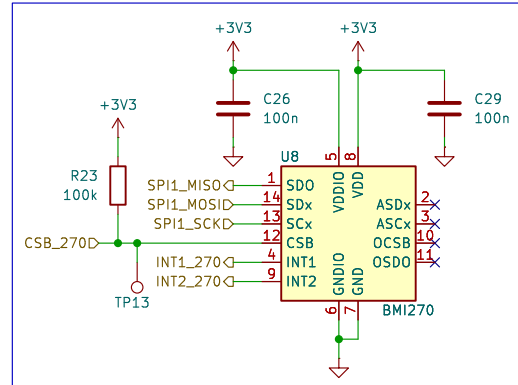
MAX30102 (HR/SPO2) + LEVEL-SHIFTING



BME688 (AQ, VOC, TEMP, PRES, HUM)

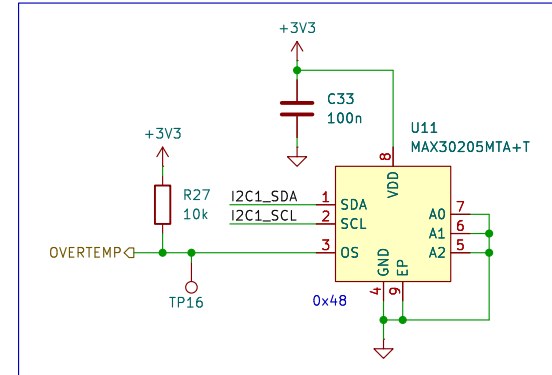


BMI 270 (GYRO & ACCEL)



**270 NOT RECOMMENDED FOR
NEW DESIGNS. USE BMI323**

MAX30205 BODY TEMP



-NEED ESD PROTECTION ON ALL EXTERNAL FACING CONNECTORS
-ESD NEEDED ON BOARD TO BOARD INTERCONNECTS FOR GPS MOD
-TERMINATION NOT NECESSARY
-OPEN DRAIN PINS ARE 3V3 TOLERANT
VERIFIED AFTER TESTING
TEST: CODE 836-30 DATE: 2/19/26

**ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC**

Sheet: /SENSORS/
File: SENSORS.kicad_sch

Title: SENSORS

Size: A4 Date: 2026-01-14

KiCad E.D.A. 9.0.7

Rev: 1.0.0

Id: 4/4